

Abdullah Hashmat

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EDUCATION

Lahore University of Management Science - BS Computer Science

Aug 2021-May 2025

Relevant Courses: CS6304 Advanced Machine Learning, CS5302 Speech Processing with Generative AI, CS6303 Large Language Models, CS331 Artificial Intelligence, CS487 Cloud Development, CS437 Deep Learning

TECHNICAL SKILLS & INTERESTS

Languages: English, Urdu

Technical Skills: Python, MERN, Java, C++/C, MATLAB, Haskell, React, LaTeX, PyTorch, AWS

Interests: Formula 1, Golf, Horse Riding, Applied Machine Learning, Data Science

RESEARCH EXPERIENCE

Max Planck Institute SWS

April 2026 - Present

Research Intern | Supervised by Dr. Abhilasha Ravichander

- Designing an LLM driven agentic system for inverse molecular design, combining generative models with structured tool-calling (RDKit, docking, QSAR) for multi-objective molecular optimization
- Building closed loop generation pipelines with iterative refinement and feedback driven reasoning, enabling improved validity, diversity, and binding affinity across candidate molecules

Center for Speech and Language Technologies (CSaLT) – LUMS

Sept 2024 – Present

Researcher | Supervised by Dr. Agha Ali Raza (LUMS) & Dr. Usman Naseem

- Developing self-evolving agentic frameworks with built-in cultural awareness, enhancing reasoning and tool-calling performance while improving robustness to Western-centric alignment biases
- Published Pakistan's first locally contextualized QA bias benchmark (PakBBQ) (17,180 QA pairs; 8 socio-cultural bias categories), benchmarking GPT-4.1, Gemini 2.0, and DeepSeek-V3. Demonstrated cross-linguistic bias disparities and up to ~12-point performance shifts.
- Led a team and developed Pak3H, a human validated Urdu benchmark suite for evaluating LLM alignment (helpfulness, harmlessness, honesty), exposing significant cross-lingual degradation in safety and truthfulness
- Researched 10+ Large Language Models (LLMs) for low-resource settings, analyzing bias vulnerabilities and jailbreak robustness, while optimizing speech/NLP models for Urdu language tasks.

German Research Center for AI & RPTU

July 2025 – March 2026

Lead Researcher | Supervised by Dr. Nabeel Asim (RPTU, DFKI)

- Developed a multi representation SMILES based Molecular Property Prediction Model using a multi-granular transformer framework leveraging atom-, functional-group-, and branch-level SMILES representations for robust molecular property prediction.
- Pretrained specialized **BERT**, **RoBERTa**, **DeBERTa** and **statespace** models on large scale SMILES corpora and distilled their knowledge into a unified student model using MLM + FitNet distillation with dynamic gating.
- Achieved **state-of-the-art** and consistent performance across **all 22 ADMET tasks**, outperforming models that excel only on task subsets.
- Built and processed a **500M+** compound dataset across **150+** molecular properties, leveraging large scale data aggregation, web scraping, and distributed GPU clusters for high throughput processing

Confidence Aware Multi-Teacher Distillation for OOD Generalization (CITY, LUMS) [Link](#)

Sept 2024 – Mar 2025

Primary Researcher | Supervised by Dr. Muhammad Tahir (LUMS)

- Proposed a confidence-aware multi-teacher knowledge distillation framework to mitigate texture/shape inductive biases and improve out-of-distribution (OOD) generalization in image classification useful in medical imaging
- Ensembled shape- and texture-biased ViT and VGG16 teachers using entropy-based logit weighting; trained a ViT-Small student via KL divergence with dynamic bias propagation.
- Achieved balanced cue reliance (Shape 0.49% / Texture 0.51%) and outperformed single-teacher and naïve KD baselines across ImageNet-1k, Stylized-ImageNet, and related OOD benchmarks, demonstrating improved robustness.

RESEARCH PAPERS & PUBLICATIONS

IRTIQA: Evolving in English, Evaluated in Urdu — Does Agent Self-Evolution Transfer Across Languages?—

(Under ARR Review) [Link](#)

April 2026 – Aug 2026

- Developed IRTIQA, a multilingual benchmark for evaluating self-evolving agents across English, Urdu, and code-switching.
- Found persistent 12–26 point multilingual gaps, especially on reasoning and longer horizon agent tasks.

Pak3H- Evaluating the Cost of Cultural Mismatch in LLM Alignment (EMNLP Main 2026) [Link](#)

Jan 2026 – March 2026

- Developed Pak3H, a human-validated Urdu benchmark suite of total 28000 samples for evaluating LLM alignment (helpfulness, harmlessness, honesty) in culturally grounded settings
- Designed a multi-stage annotation and localization pipeline ensuring semantic and cultural fidelity beyond translation based approaches
- Revealed systematic cross lingual alignment failures in state of the art LLMs, with consistent degradation across all 3H dimensions

PakBBQ-A Culturally Adapted Bias Benchmark for QA (EMNLP Main 2025) [Paper Link](#)

Sept 2024 – May 2025

- Developed PakBBQ, the first culturally adapted QA benchmark (17,180 examples in Urdu and English across 8 sociocultural bias categories) to evaluate social bias in multilingual LLMs within the Pakistani context

- Evaluated 6 multilingual LLMs under zero shot conditions, revealing cross linguistic bias gaps and demonstrating that disambiguation and negative framing significantly improved bias mitigation and accuracy

TEACHING ASSISTANTSHIPS

- **Teaching Assistant** – *AI600 Machine Learning (Masters in AI)* Jan 2025 – May 2025
- **Teaching Assistant** – *CS5302 Foundations of Generative AI* Jan 2025 – May 2025
- **Teaching Assistant** – *CS535 Machine Learning* Sept 2024 – Dec 2024
- **Teaching Assistant** – *CS100 Computational Problem Solving* Sept 2023 – Dec 2023

GITHUB PROJECTS

- Bias Mitigation in Deep Learning Vision Models** [Link](#) Sept 2024 – Dec 2024
 - Implemented a multi-teacher single-student knowledge distillation framework to address inductive biases in deep learning, leveraging CNNs for texture and Vision Transformers for shape representation
 - Designed space adapters for heterogeneous architectures, achieving a compact student model with superior generalisation on cue-conflict datasets under distributional shifts, resulting in **12%** less bias.
- AI for All – Evaluating Transformer and LLM Models in Low Resourced Settings** [Link](#) Sept 2024 – Dec 2024
 - Evaluated open source LLM and transformer model’s performance on low resource languages such as Urdu, Sindhi and Pashto. Tested jail breaking adversarial prompts with a success rate of up to **40%** in Llama-3B.
- AI-Driven Stock Insights Platform with AWS & Claude Chatbot** [Link](#) Sept 2024 – Dec 2024
 - Built and deployed full-stack real-time stock analysis platform using AWS Lambda, S3, ECS, and Neon Postgres
 - Integrated Claude Haiku based chatbot with custom prompt chains for natural language queries and investment insights
 - Achieved **<200ms** latency and auto-scaling to millions of users via serverless architecture and ECS Fargate
- Agentic Data Science Chatbot** [Link](#) Jan 2024 – May 2024
 - Built an agentic AI Data Science chatbot, aimed to perform intuitive statistical analysis, EDA and data cleaning, enhancing accessibility for users with limited Data Science knowledge
 - Fine tuned Llama 3 to analyse data in **10+** supported formats, deployed on Hugging Face using Gradio

INDUSTRY EXPERIENCE

- Darcheeni (AI in Healthcare Startup) – LUMS – MLOps Engineer** May 2024 – May 2025
 - Led backend development for an AI-driven clinical assistant, integrating ASR, RL-based decision models, and LLM-driven response generation for real-time triage
 - Optimized speech recognition pipelines through data processing, model finetuning, and LLM-based post-processing for improved transcription quality in diverse settings
 - Designed and deployed serverless infrastructure on AWS (Lambda, DynamoDB, Auth0) with autoscaling, enabling scalable, low-latency model inference and system reliability
- Delivery Hero (FoodPanda) – Sales Performance Intern** Jun 2024 – Aug 2024
 - Conducted large-scale EDA on vendor onboarding and QC data to identify trends and bottlenecks, streamlining the self-signup process and reducing QC rejection rate by **15%**.
 - Developed an agentic workflow using Python and Google APIs to automatically pull daily data, generate invoices, and distribute them to vendors via WhatsApp, reducing manual effort and process time by **35%**.
- Pakistan Television Network – Data Science Intern** Jun 2023 – Jul 2023
 - Revamped and enhanced PTV World’s ticker headline system using machine learning models for text similarity, resulting in **20%** reduction in headline overheads
 - Built a predictive analysis model, enhancing viewer engagement metrics through data-driven content strategies by **10%**

LEADERSHIP EXPERIENCE AND CO-CURRICULAR ACTIVITIES

- TechStars Startup Weekend, Lahore - Convenor** Jul 2022 – Mar 2023
 - Executed Startup Weekend Lahore 2023 in collaboration with National Incubation Center Lahore and Google for Startups, led workshops and partnered with top industry leaders for startup guidance and judging
 - Managed a 54-hour event for over **40** startups and **150+** attendees across Pakistan, overlooking a team of 60 and securing **800k+** PKR in sponsorships
- SPADES - Assistant Director (Socials and Operations)** Sept 2022 – May 2023
 - Led the organisation of PSIFT's flagship event, coordinating social function and concerts for over **1,400** attendees